Cell Tuition
1-1 PRIVATE TUTORS

NUMBER

Ratios

Name _____ Date _____ Calculator allowed Version 1

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KEY WORDS ratio *(ōwehenga)* · simplest form *(āhua māmā rawa)* · equivalent *(ōrite)* · share · part

A Simplifying ratios ●○○

Remember: Divide every part of the ratio by the same number. Change to the same units first.

1. Write each ratio in simplest form.

a) 6 : 9 _____ **b)** 10 : 25 _____ **c)** 24 : 16 _____ **d)** 35 : 21 _____

e) 45 : 60 _____ **f)** 18 : 30 : 12 _____ **g)** 20 cm : 1 m _____ **h)** 50c : $2 _____

B Equivalent ratios ●○○

2. Find the missing number.

a) $2 : 3 = 8 : \square$ _____ **b)** $5 : 4 = \square : 20$ _____ **c)** $3 : 7 = 15 : \square$ _____

d) $\square : 6 = 20 : 24$ _____ **e)** $3 : \square = 9 : 12$ _____ **f)** $4 : \square = 28 : 35$ _____

C Sharing in a ratio ●●○

Remember: Add the parts of the ratio. Divide the amount by that total to find one part.

3. Share each amount in the ratio given.

a) Share $40 in the ratio 3 : 5.

b) Share 60 in the ratio 1 : 4.

c) Share $72 in the ratio 5 : 7.

d) Share 180 in the ratio 2 : 3 : 4.

e) Share 350 g in the ratio 3 : 2.

f) Share $1,200 in the ratio 7 : 5.

g) Share 96 in the ratio 1 : 2 : 5.

h) Share 4.5 L in the ratio 4 : 5.

D Finding one part from another ●●○

Watch out: Check which part of the ratio the number you are given belongs to.

4. Answer each question.

a) The ratio of boys to girls in a club is 3 : 4. There are 12 boys. How many girls are there?

b) A cake uses flour and sugar in the ratio 5 : 2. How much sugar goes with 300 g of flour?

c) Red and blue paint are mixed in the ratio 2 : 7. How much red paint is needed for 14 L of blue?

d) On a school trip the ratio of adults to children is 1 : 6. There are 42 children. How many adults are there?

e) Ana and Ben share some money in the ratio 3 : 5. Ana gets $24. How much do they share altogether?

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E Ratios and fractions ●●○

5. Answer each question.

a) A bag has red and blue counters in the ratio 2 : 3. What fraction of the counters are red? _____

b) A class has boys and girls in the ratio 5 : 4. What fraction of the class are girls? _____

c) $\frac{3}{7}$ of the players in a team are left-handed. Write the ratio of left-handed to right-handed players. _____

d) $\frac{5}{8}$ of a drink is water and the rest is juice. Write the ratio of water to juice. _____

F Proportion and scale ●●○

> **Remember:** Find the value of one first, then multiply.

6. Answer each question.

a) 5 pens cost $8.50. Find the cost of 8 pens.

b) 3 kg of apples cost $11.70. Find the cost of 5 kg.

c) A car travels 150 km on 12 litres of petrol. How far does it travel on 20 litres?

d) A house plan has a scale of 1 : 200. A room is 4.5 cm long on the plan. Find its real length in metres.

G Problems ●●●

7. Juice is made by mixing concentrate and water in the ratio 1 : 4. How much of each is needed to make 2 litres (2,000 mL) of juice?

8. The ages of Mia and Tom are in the ratio 3 : 5. Tom is 20. What will the ratio of their ages be in 4 years? Give it in simplest form.

9. Concrete is made from cement, sand and gravel in the ratio 1 : 2 : 4. How much of each is needed to make 350 kg of concrete?

10. A map has a scale of 1 : 50,000. Two towns are 8 cm apart on the map. How far apart are they in real life? Give your answer in kilometres.

11. Two numbers are in the ratio 5 : 3. Their difference is 18. Find the two numbers.

12. The angles of a triangle are in the ratio 2 : 3 : 4. Find the largest angle.

13. A recipe for 4 people uses 300 g of rice. How much rice is needed for 10 people?

★ Extension

14. Kiri and Tama share some money in the ratio 7 : 3. Kiri gets $36 more than Tama. How much do they share altogether?

15. The ratio of cats to dogs at a shelter is 5 : 4. Six more dogs arrive and the ratio becomes 5 : 6. How many cats are there?

16. a : b = 2 : 3 and b : c = 6 : 5. Write a : b : c in simplest form.

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Cell Tuition
1-1 PRIVATE TUTORS

NUMBER

Ratios

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS ratio *(ōwehenga)* · simplest form *(āhua māmā rawa)* · equivalent *(ōrite)* · share · part

A Simplifying ratios ●○○

Remember: Divide every part of the ratio by the same number. Change to the same units first.

1. Write each ratio in simplest form.

a) 8 : 12 _____ **b)** 15 : 35 _____ **c)** 27 : 18 _____ **d)** 28 : 20 _____

e) 36 : 48 _____ **f)** 24 : 16 : 40 _____ **g)** 25 cm : 1 m _____ **h)** 40c : $3 _____

B Equivalent ratios ●○○

2. Find the missing number.

a) $3 : 4 = 12 : \square$ _____ **b)** $7 : 2 = \square : 10$ _____ **c)** $4 : 9 = 20 : \square$ _____

d) $\square : 8 = 18 : 48$ _____ **e)** $5 : \square = 15 : 21$ _____ **f)** $6 : \square = 42 : 35$ _____

C Sharing in a ratio ●●○

Remember: Add the parts of the ratio. Divide the amount by that total to find one part.

3. Share each amount in the ratio given.

a) Share $56 in the ratio 3 : 4.

b) Share 75 in the ratio 1 : 4.

c) Share $84 in the ratio 5 : 9.

d) Share 240 in the ratio 1 : 3 : 4.

e) Share 450 g in the ratio 4 : 5.

f) Share $1,500 in the ratio 7 : 8.

g) Share 108 in the ratio 2 : 3 : 4.

h) Share 3.5 L in the ratio 3 : 4.

D Finding one part from another ●●○

Watch out: Check which part of the ratio the number you are given belongs to.

4. Answer each question.

a) The ratio of boys to girls in a club is 4 : 5. There are 20 boys. How many girls are there?

b) A cake uses flour and sugar in the ratio 7 : 3. How much sugar goes with 350 g of flour?

c) Red and blue paint are mixed in the ratio 3 : 8. How much red paint is needed for 24 L of blue?

d) On a school trip the ratio of adults to children is 1 : 8. There are 56 children. How many adults are there?

e) Ana and Ben share some money in the ratio 2 : 7. Ana gets $18. How much do they share altogether?

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E Ratios and fractions ●●○

5. Answer each question.

a) A bag has red and blue counters in the ratio 3 : 5. What fraction of the counters are red? _____

b) A class has boys and girls in the ratio 4 : 7. What fraction of the class are girls? _____

c) $\frac{2}{9}$ of the players in a team are left-handed. Write the ratio of left-handed to right-handed players. _____

d) $\frac{3}{5}$ of a drink is water and the rest is juice. Write the ratio of water to juice. _____

F Proportion and scale ●●○

> **Remember:** Find the value of one first, then multiply.

6. Answer each question.

a) 4 pens cost $7.60. Find the cost of 7 pens.

b) 2 kg of apples cost $8.40. Find the cost of 5 kg.

c) A car travels 180 km on 15 litres of petrol. How far does it travel on 25 litres?

d) A house plan has a scale of 1 : 200. A room is 3.5 cm long on the plan. Find its real length in metres.

G Problems ●●●

7. Juice is made by mixing concentrate and water in the ratio 1 : 5. How much of each is needed to make 1.5 litres (1,500 mL) of juice?

8. The ages of Mia and Tom are in the ratio 2 : 3. Tom is 18. What will the ratio of their ages be in 6 years? Give it in simplest form.

9. Concrete is made from cement, sand and gravel in the ratio 1 : 2 : 3. How much of each is needed to make 420 kg of concrete?

10. A map has a scale of 1 : 25,000. Two towns are 12 cm apart on the map. How far apart are they in real life? Give your answer in kilometres.

11. Two numbers are in the ratio 7 : 4. Their difference is 21. Find the two numbers.

12. The angles of a triangle are in the ratio 1 : 3 : 5. Find the largest angle.

13. A recipe for 6 people uses 450 g of rice. How much rice is needed for 10 people?

★ Extension

14. Kiri and Tama share some money in the ratio 8 : 3. Kiri gets $45 more than Tama. How much do they share altogether?

15. The ratio of cats to dogs at a shelter is 4 : 3. Eight more dogs arrive and the ratio becomes 4 : 5. How many cats are there?

16. a : b = 3 : 4 and b : c = 8 : 5. Write a : b : c in simplest form.

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